

Electric Fields

Michael Faraday



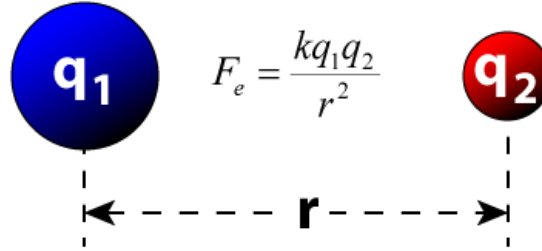
Born	22 September 1791 Newington Butts, England
Died	25 August 1867 (aged 75) Hampton Court, Middlesex, England
Residence	United Kingdom
Nationality	British
Known for	Faraday's law of induction Electrochemistry Faraday effect Faraday cage Faraday constant Faraday cup Faraday's laws of electrolysis Faraday paradox Faraday rotator Faraday-efficiency effect Faraday wave Faraday wheel Lines of force



Michael Faraday FRS was an English scientist who contributed to the study of electromagnetism and electrochemistry. His main discoveries include the principles underlying electromagnetic induction, diamagnetism and electrolysis.

Some “Simple” Questions...

Coulomb's law:



How do q_1 and q_2 “know” of each other’s presence?

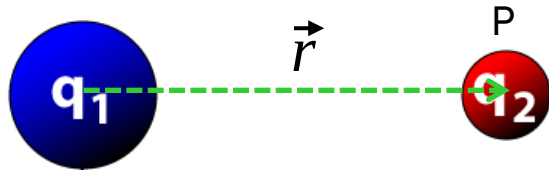
How can they exert forces without touching each other?

What will happen to the force if the distance changes?

What if the two charges are sitting across our galaxy?

.....

Action over Distance or Electric Field?

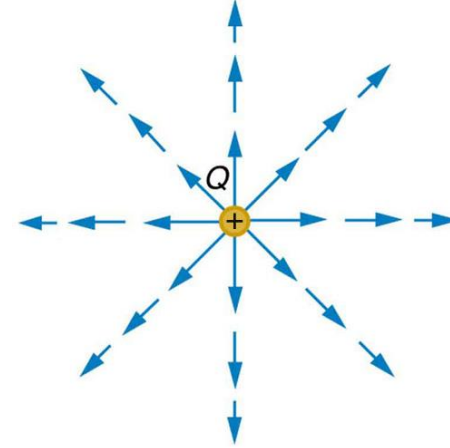
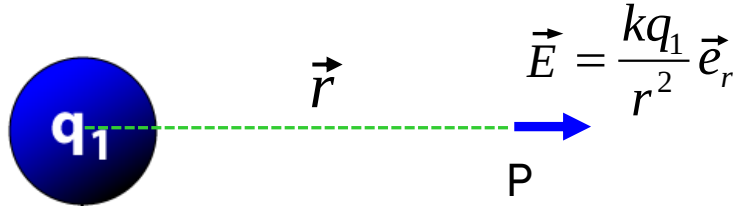


$$\vec{F}_e = \frac{kq_1q_2}{r^2} \vec{e}_r = \frac{kq_1}{r^2} \vec{e}_r \cdot q_2 = \vec{E} \cdot q_2$$

Instead of saying that q_1 and q_2 interact with each other over a distance, we now view the situation in a different way:

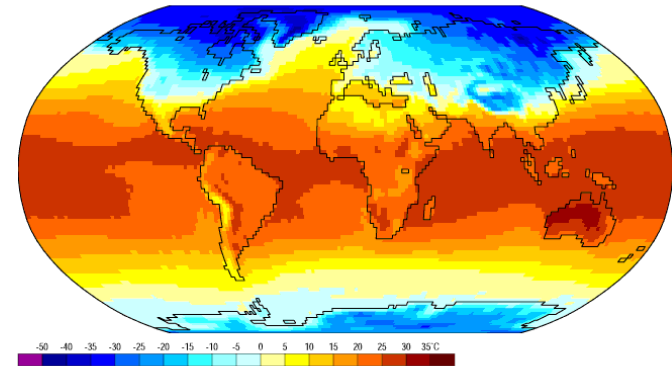
- ❖ q_1 sets up an **electric field** $\vec{E}$ in the space surrounding it, and q_2 “feels” such a field.
- ❖ At any given point P in space, the electric field has both magnitude and direction. The magnitude depends on the magnitude of q_1 and the distance between P and q_1 . The direction depends on the direction from q_1 to P **and the sign of q_1** .
- ❖ **What is the electric field created by q_2 ? How does q_1 feel this field?**

Electric Field

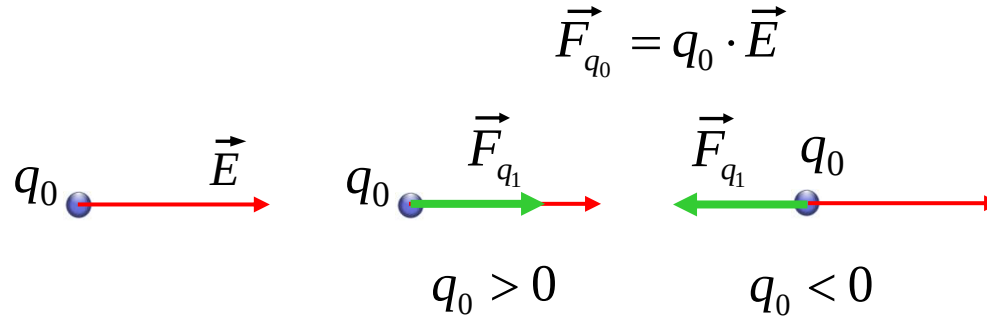


- ❖ A field is a physical quantity that has a value at each point in space.
- ❖ A field can be classified as a scalar field or a vector field, depending on whether its value at each point is a scalar or a vector. **Electric field is a vector field.**

A scalar field: mean January air temperature of the Earth's surface



Particle in Electric Field



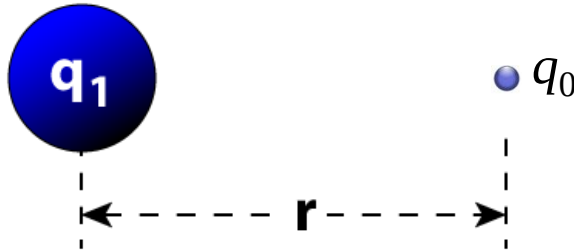
- ❖ If we know the electric field at a given point, the force that a test charge q_0 feels at that point is:

$$\vec{F} = q \cdot \vec{E}$$

- ❖ Direction of the force:
 - positive charges – in the direction of the field
 - negative charges – in the opposite direction of the field

Electric Field of Point Charge

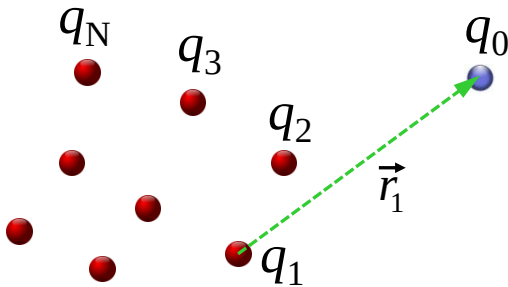
Single charge



$$\vec{F} = \frac{kq_1q_0}{r^2} \vec{e}_r = q_0 \cdot \vec{E}$$

$$\vec{E} = \frac{kq_1}{r^2} \vec{e}_r$$

Multiple charges



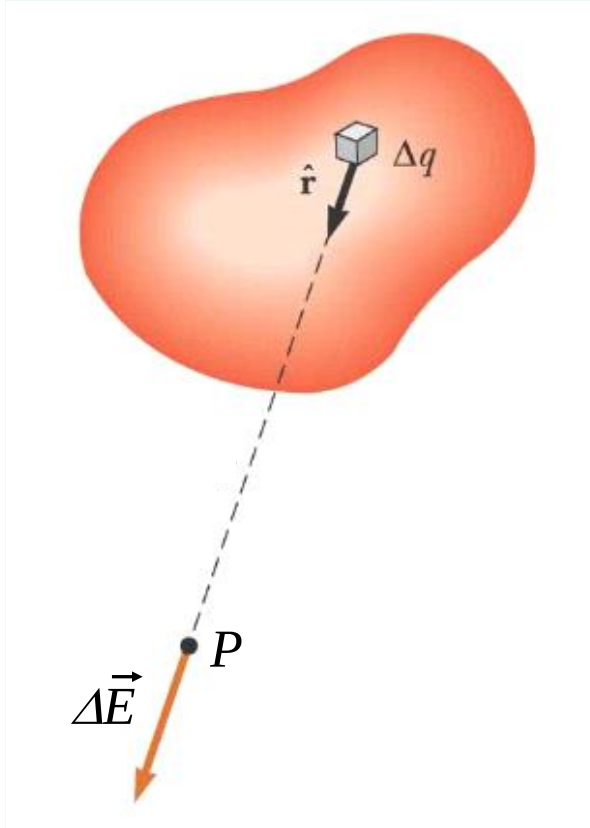
$$\vec{F}_0 = \vec{F}_{0,1} + \vec{F}_{0,2} + \dots + \vec{F}_{0,N}$$

$$\vec{E} = \frac{\vec{F}}{q_0} = \frac{\vec{F}_{0,1} + \vec{F}_{0,2} + \dots + \vec{F}_{0,N}}{q_0}$$

$$= \vec{E}_1 + \vec{E}_2 + \dots + \vec{E}_N$$

$$= \frac{kq_1}{r_1^2} \vec{e}_{r_1} + \frac{kq_2}{r_2^2} \vec{e}_{r_2} + \dots + \frac{kq_N}{r_N^2} \vec{e}_{r_N}$$

Electric Field of Continuous Charge Distribution



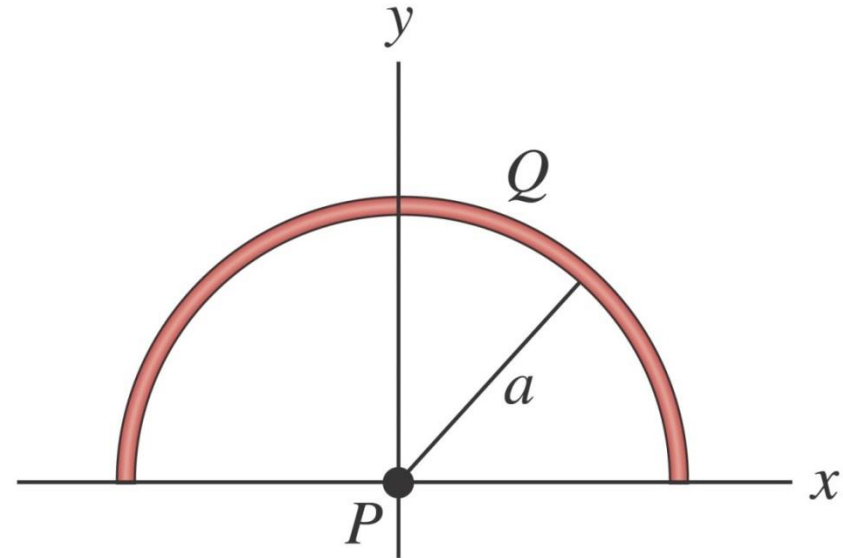
$$\Delta \vec{E} = \frac{k \Delta q}{r^2} \vec{e}_r$$

$$\vec{E} = \sum \Delta \vec{E} \rightarrow \int d\vec{E} = k \int \frac{dq}{r^2} \vec{e}_r$$

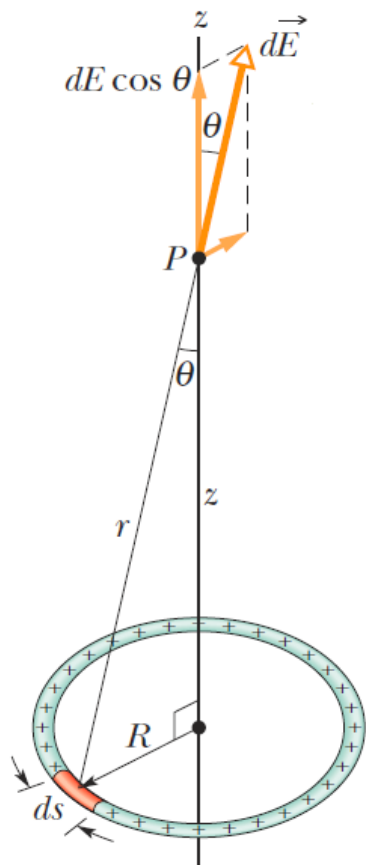
Example Problem

Positive charges (total amount of Q) are uniformly distributed on a semicircle. The electric field that these charges produce at the center of curvature P is in

- A. the $+x$ -direction.
- B. the $-x$ -direction.
- C. the $+y$ -direction.
- ✓ D. the $-y$ -direction.
- E. none of the above



Electric Field of Charged Ring



Problem: Find the electric field at the central axis of a uniformly charged ring. Parameters are shown in figure.

For a differential section ds , the charge contained is

$$dq = \lambda ds = \frac{q}{2\pi R} ds$$

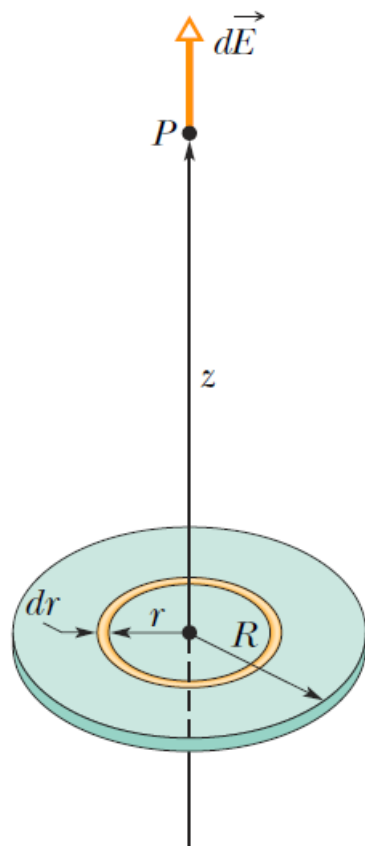
The electric field generated by ds at point $P(0, 0, z)$ is

$$\begin{aligned} d\mathbf{E} &= \frac{1}{4\pi\epsilon_0} \frac{dq}{z^2 + R^2} \mathbf{e}_R \\ &= \frac{\lambda ds}{4\pi\epsilon_0} \left[\frac{\sin\theta \mathbf{e}_x}{z^2 + R^2} + \frac{\cos\theta \mathbf{e}_z}{z^2 + R^2} \right] \end{aligned}$$

Since components along $\mathbf{e}_x$ cancels out due to symmetry, we have

$$\begin{aligned} \mathbf{E} &= \int d\mathbf{E} = \frac{\lambda}{4\pi\epsilon_0} \int ds \frac{\cos\theta \mathbf{e}_z}{z^2 + R^2} \quad \text{Const. Take out from integral!} \\ &= \frac{\lambda}{4\pi\epsilon_0} \frac{z \mathbf{e}_z}{(z^2 + R^2)^{3/2}} \int ds \\ &= \frac{\lambda}{4\pi\epsilon_0} \frac{z \mathbf{e}_z}{(z^2 + R^2)^{3/2}} 2\pi R = \frac{qz \mathbf{e}_z}{4\pi\epsilon_0 (z^2 + R^2)^{3/2}} \end{aligned}$$

Electric Field of Charged Disk



Problem: Find the electric field at the central axis of a uniformly charged disk. Parameters are shown in figure.

Consider a differential **ring** with radius $r < R$, with charge

$$dq = \sigma dA = \frac{q}{\pi R^2} 2\pi r dr$$

According to our previous result, the field generated by this ring is

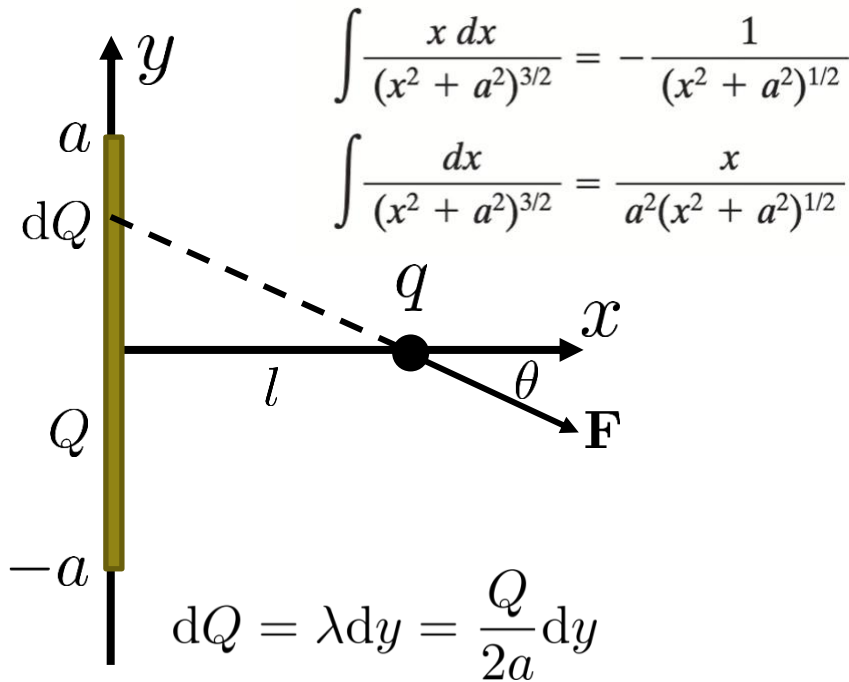
$$d\mathbf{E} = dq \frac{z\mathbf{e}_z}{4\pi\epsilon_0(z^2 + r^2)^{3/2}} = \frac{\sigma z\mathbf{e}_z}{2\epsilon_0} \frac{r dr}{(z^2 + r^2)^{3/2}}$$

Therefore, the total field at $P(0, 0, z)$ is

$$\begin{aligned} \mathbf{E} &= \int d\mathbf{E} = \frac{\sigma z\mathbf{e}_z}{2\epsilon_0} \int_0^R \frac{r dr}{(z^2 + r^2)^{3/2}} \\ &= \frac{\sigma z\mathbf{e}_z}{4\epsilon_0} \int_0^R \frac{d(r^2)}{(z^2 + r^2)^{3/2}} = \frac{\sigma z\mathbf{e}_z}{4\epsilon_0} \int_0^R \frac{d(r^2 + z^2)}{(z^2 + r^2)^{3/2}} \\ &= \frac{\sigma z\mathbf{e}_z}{4\epsilon_0} \left[-\frac{2}{\sqrt{z^2 + r^2}} \right]_0^R = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right) \mathbf{e}_z \end{aligned}$$

A More Difficult Problem

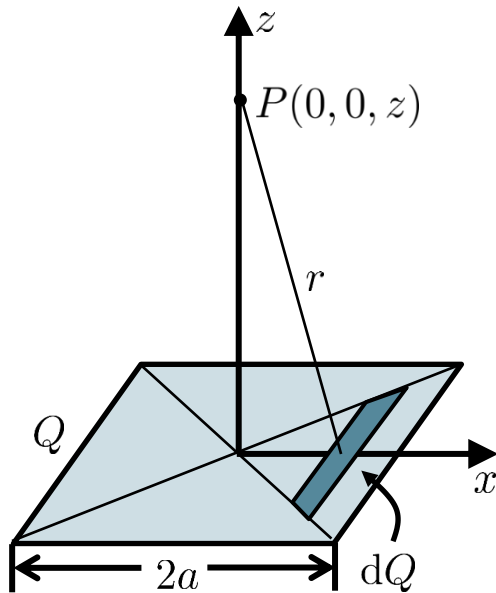
Example: A rod with length $2a$ and uniformly distributed charge Q is placed on the y -axis, spanning the region $\{(x, y) | x = 0, -a < y < a\}$. Find the electric force felt by a charge q at point $(l, 0)$.



$$\begin{aligned} F_x &= \int_{-a}^a \frac{1}{4\pi\epsilon_0} \frac{q dQ}{l^2 + y^2} \cos \theta \\ &= \frac{1}{4\pi\epsilon_0} \frac{Qq}{2a} \int_{-a}^a \frac{dy}{l^2 + y^2} \frac{l}{\sqrt{l^2 + y^2}} \\ &= \frac{Qql}{8\pi\epsilon_0 a} \frac{2a}{l^2 \sqrt{a^2 + l^2}} \\ &= \frac{Qq}{4\pi\epsilon_0 l \sqrt{a^2 + l^2}}; \\ F_y &= \int_{-a}^a \frac{1}{4\pi\epsilon_0} \frac{q dQ}{l^2 + y^2} \sin \theta = 0 \\ \text{So, } \mathbf{F} &= \frac{Qq}{4\pi\epsilon_0 l \sqrt{a^2 + l^2}} \mathbf{e}_x \end{aligned}$$

Electric Field of Charged Disk

Problem: Find the electric field at the central axis of a uniformly charged **squared** disk. Parameters are shown in figure.



Consider a linear charged stripe dQ as shown. From an example in Chapter 21 (refer to page 35 of the ppt for Chapter 21), we have the field generated by dQ at P :

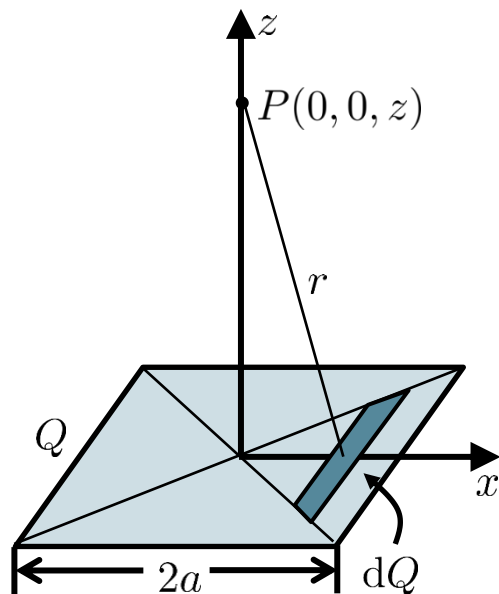
$$d\mathbf{E} = \frac{dQ}{4\pi\epsilon_0 r \sqrt{x^2 + r^2}} \mathbf{e}_r$$

$$\text{where } dQ = \sigma 2x dx \text{ and } \sigma = \frac{Q}{4a^2}$$

Electric Field of Charged Disk

$$d\mathbf{E} = \frac{dQ}{4\pi\epsilon_0 r \sqrt{x^2 + r^2}} \mathbf{e}_r$$

$$dQ = \sigma 2x dx.$$



Since r depends on x , and only E_z is important,

$$dE_z = \frac{\sigma x dx}{2\pi\epsilon_0} \frac{z}{(x^2 + z^2)\sqrt{2x^2 + z^2}}$$

For one quarter of the square,

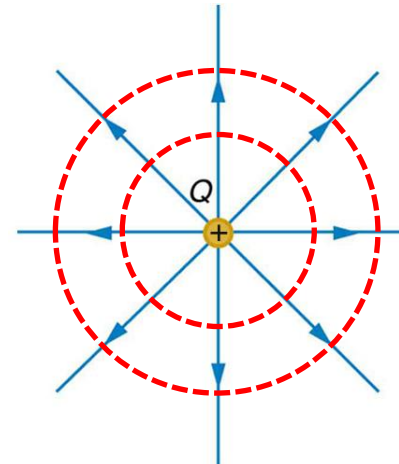
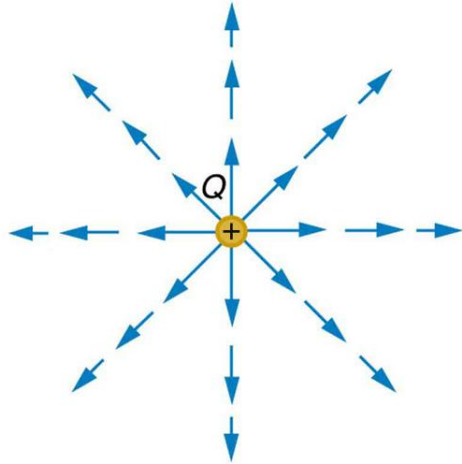
$$\begin{aligned} E_z &= \frac{\sigma z}{2\pi\epsilon_0} \int_0^a \frac{x dx}{(x^2 + z^2)\sqrt{2x^2 + z^2}} \\ &= \frac{\sigma}{2\pi\epsilon_0} \left(\arctan \sqrt{2 \left(\frac{a}{z} \right)^2 + 1} - \frac{\pi}{4} \right) \end{aligned}$$

Thus the total field at P is

$$\mathbf{E} = 4E_z \mathbf{e}_z = \frac{2\sigma}{\pi\epsilon_0} \left(\arctan \sqrt{2 \left(\frac{a}{z} \right)^2 + 1} - \frac{\pi}{4} \right) \mathbf{e}_z$$

<https://www.zhihu.com/question/411338834>

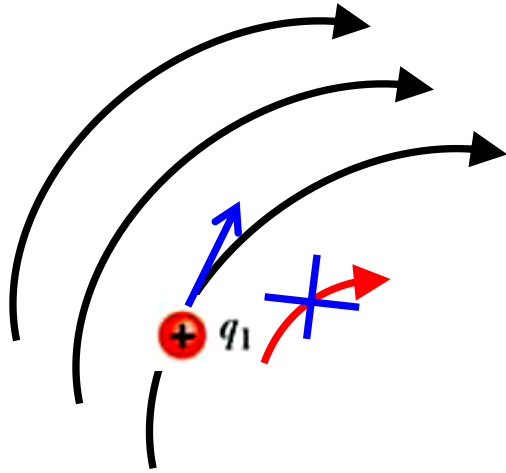
Electric Field Lines of Point Charge



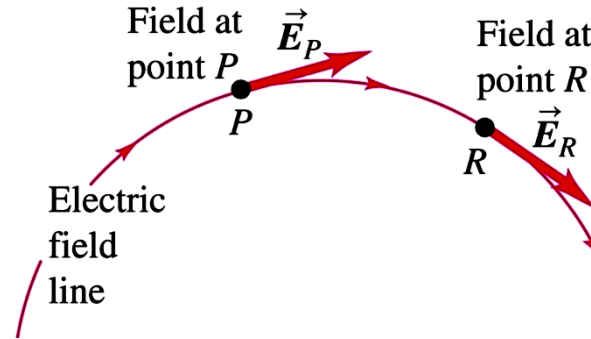
$$|\vec{E}| = \frac{kQ}{r^2}$$

- ❖ At any point, the direction of a straight field line gives the direction of the electric field at that point.
- ❖ The field lines are drawn so that the number of lines per unit area is proportional to the magnitude of the electric field.
- ❖ No field lines cross. **Why?**
- ❖ Electric field exists at every point, not only at the locations with the electric field lines shown.

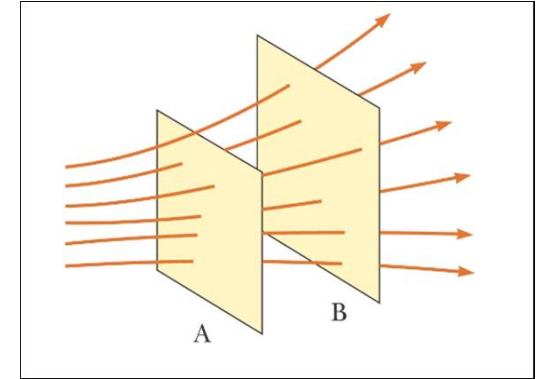
Electric Field Lines in General



Force at this point?

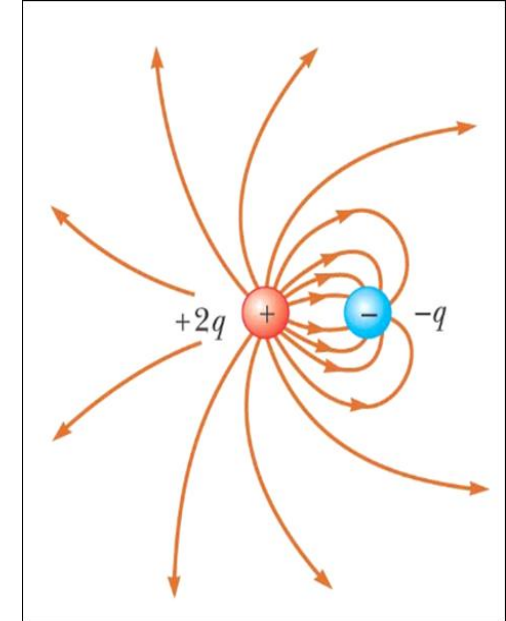
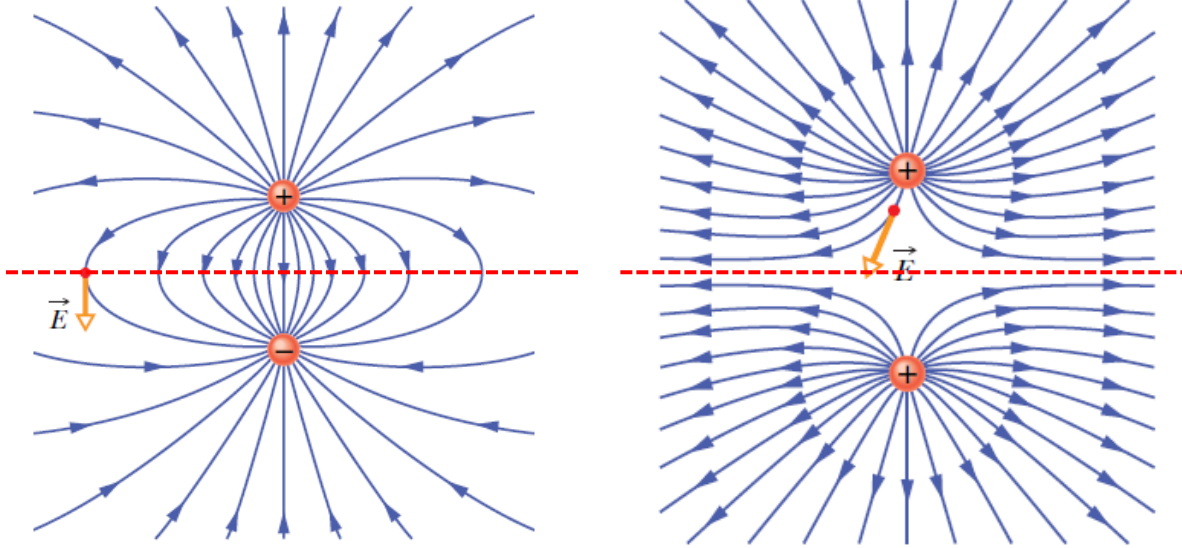


What about the magnitude?



- ❖ In general, electric field lines can be curvy!
- ❖ At any point, tangent of electric field line represents the **direction** of the electric field at that point, and density of field lines (number of lines per unit area) indicates the **magnitude**.

Examples of Electric Field Lines



Try to describe the electric fields along the red dashed lines.

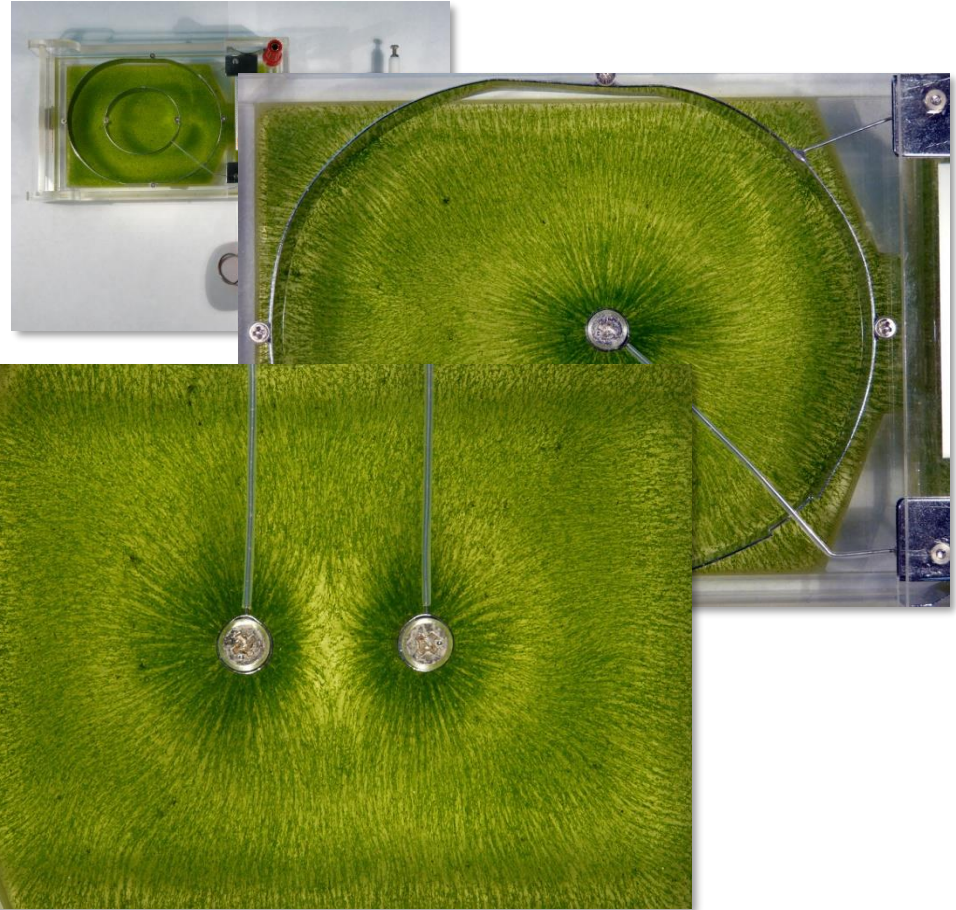
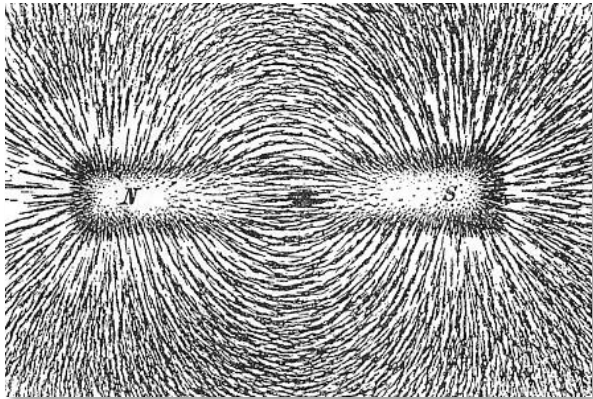
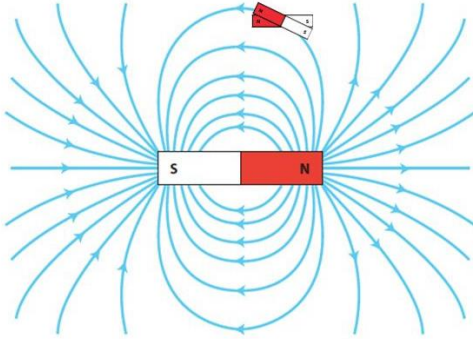
Visualizing Electric Field Lines



In the early 1800's, Michael Faraday notes that iron filings line up around a magnet to show "force field lines".

Field lines

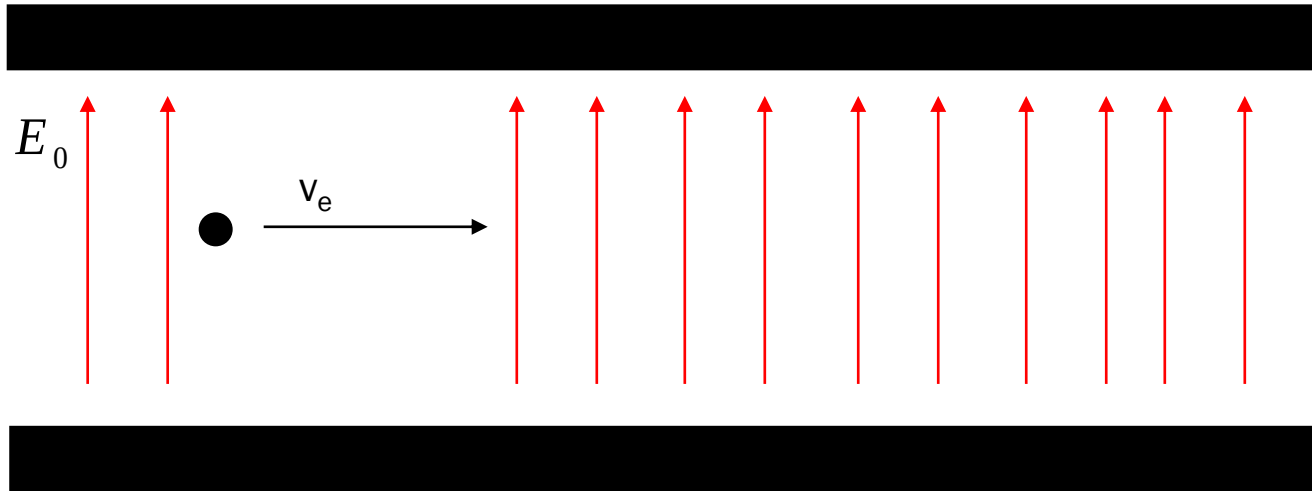
The lines show the direction of force on a north pole.



<http://tsgphysics.mit.edu/front/?page=demo.php&letnum=D%2016&show=0>

Motion of charge particles in a uniform electric field

An electron moving horizontally passes between two horizontal planes, the upper plane charged negatively, and the lower positively. A uniform, upward-directed electric field exists in this region. Describe the motion of the electron in this region.



Motion of charge particles in a uniform electric field

■ Horizontally:

- No electric field
- No force
- No acceleration
- Constant horizontal velocity

$$E_x = 0$$

$$F_x = 0$$

$$a_x = 0$$

$$v_x = v_o$$

$$x = v_o t$$

■ Vertically:

- Constant electric field
- Constant force
- Constant acceleration
- Vertical velocity increase linearly with time.

$$E_y = E_o$$

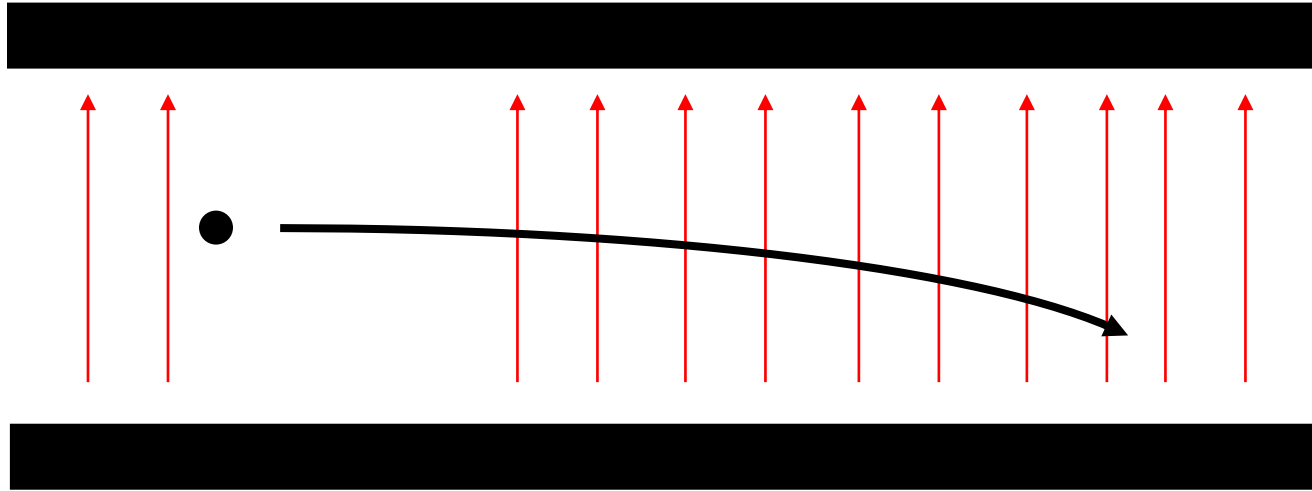
$$F_y = eE_o$$

$$a_y = eE_o / m_e$$

$$v_y = eE_o t / m_e$$

$$y = \frac{1}{2} eE_o t^2 / m_e$$

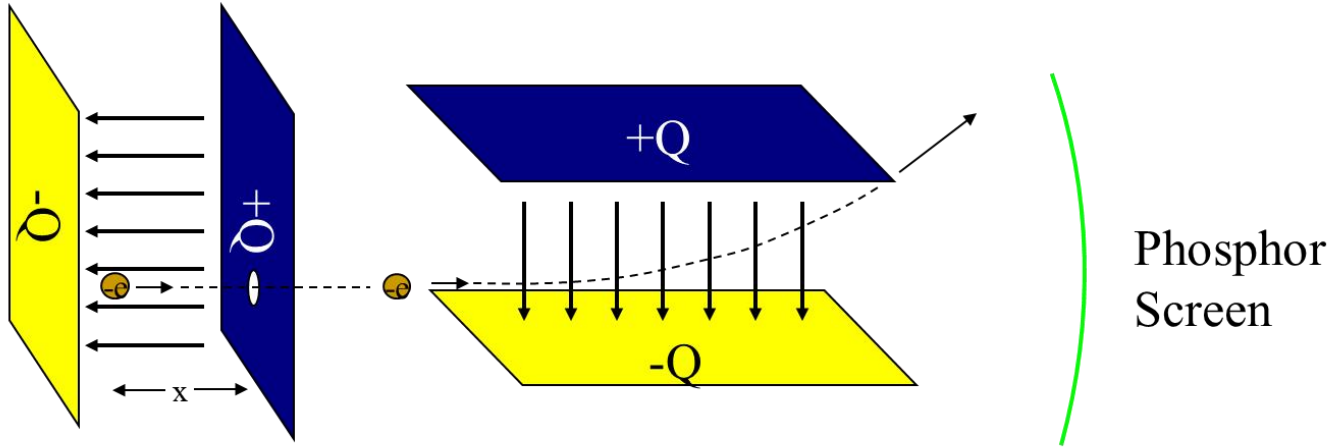
Motion of charge particles in a uniform electric field



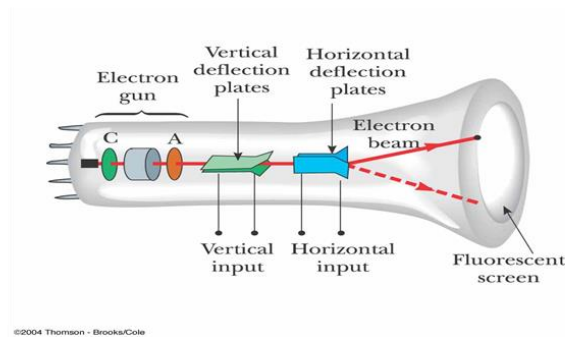
Conclusions:

- The charge will follow a parabolic path downward.
- Motion similar to motion under gravitational field only except the downward acceleration is now larger.

Cathode Ray Tube (CRT, 阴极射线管)

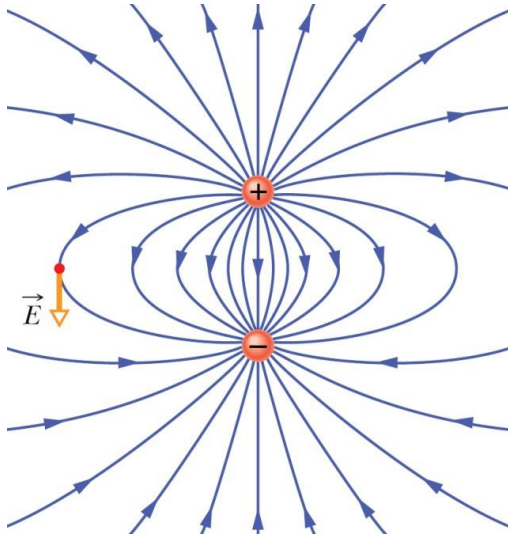


This device is known as a cathode ray tube (CRT)

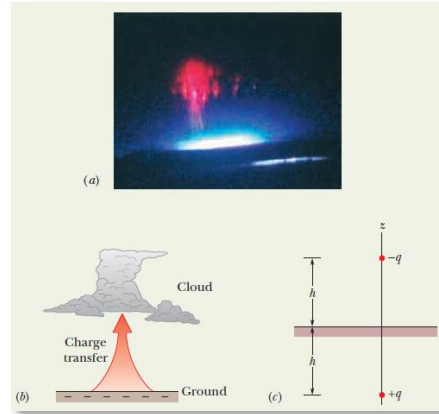


Electric Dipoles

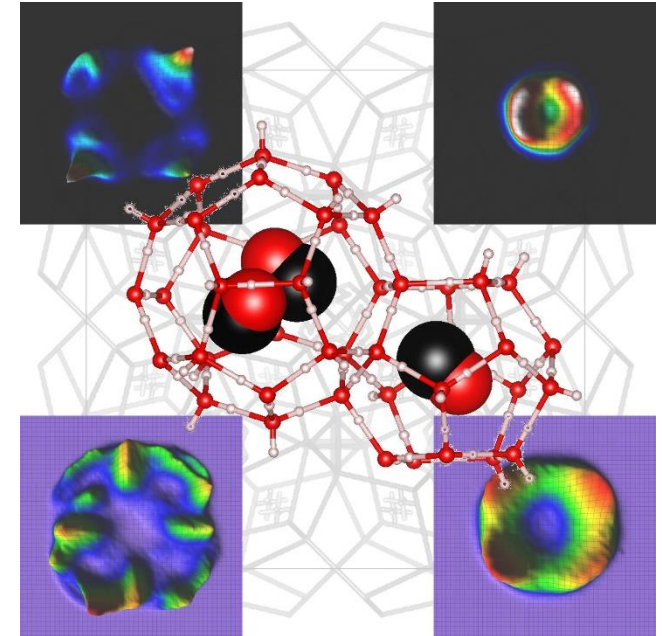
An **electric dipole** is a system of two particles that have the same charge magnitude q but opposite signs, and are separated by a distance d .



???

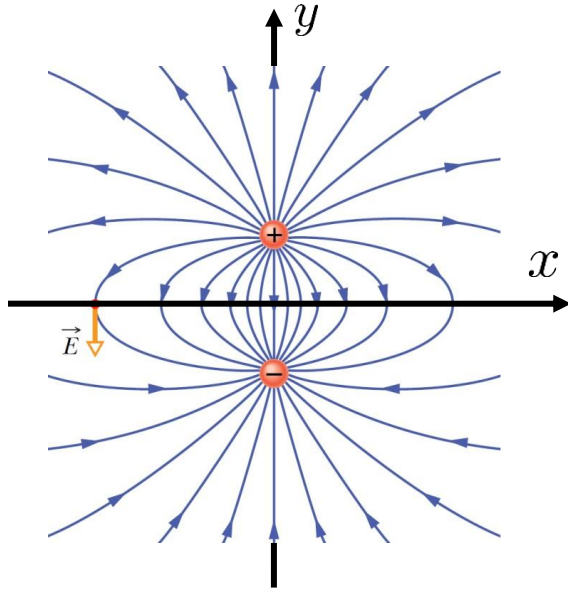


Page 565 in textbook



My own research

Electric Field of Dipole



$$\mathbf{E}(x, y) = \frac{q}{4\pi\epsilon_0} \left[\frac{x}{[x^2 + (y - d/2)^2]^{3/2}} - \frac{x}{[x^2 + (y + d/2)^2]^{3/2}} \right] \mathbf{e}_x + \frac{q}{4\pi\epsilon_0} \left[\frac{y - d/2}{[x^2 + (y - d/2)^2]^{3/2}} - \frac{y + d/2}{[x^2 + (y + d/2)^2]^{3/2}} \right] \mathbf{e}_y$$

For $x = 0$

$$\begin{aligned} \mathbf{E}(0, y) &= \frac{q}{4\pi\epsilon_0} \left[\frac{1}{(y - d/2)^2} - \frac{1}{(y + d/2)^2} \right] \mathbf{e}_y \\ &= \frac{1}{2\pi\epsilon_0 y^3} \frac{qd}{\left[1 - (d/2y)^2\right]^2} \mathbf{e}_y \end{aligned}$$

If $y \gg d$, $\mathbf{E}(0, y) = \frac{\mathbf{p}}{2\pi\epsilon_0 y^3}$, where $\mathbf{p} = qd\mathbf{e}_y$.

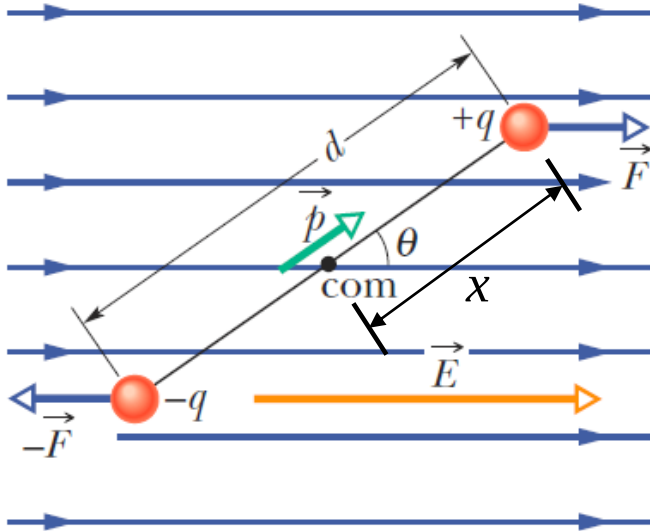
For $y = 0$

$$\mathbf{E}(x, 0) = \frac{q}{4\pi\epsilon_0} \frac{-d}{\left(x^2 + \frac{d^2}{4}\right)^{3/2}} \mathbf{e}_y = -\frac{\mathbf{p}}{4\pi\epsilon_0 \left(x^2 + \frac{d^2}{4}\right)^{3/2}}$$

If $x \gg d$, $\mathbf{E}(x, 0) = -\frac{\mathbf{p}}{4\pi\epsilon_0 x^3}$.

Dipole in Electric Field

An electric dipole has its own electric field, but here we discuss the behavior of the dipole in another given electric field. Don't mix up these two electric fields.



A dipole in a uniform E -field (see figure) produces *no net force* but a net torque τ . By definition of τ , its magnitude reads

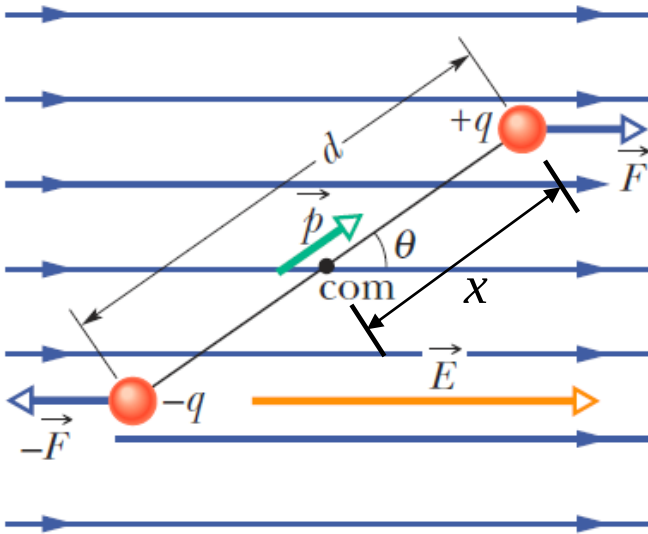
$$\begin{aligned}\tau &= (+q)Ex \sin \theta - (-q)E(d - x) \sin \theta \\ &= qEd \sin \theta,\end{aligned}$$

The direction of τ is defined by the right hand rule from $\mathbf{p}$ to $\mathbf{E}$, i.e., into the screen. Apparently, we have

$$\boxed{\boldsymbol{\tau} = \mathbf{p} \times \mathbf{E}}$$

What's the effect of such a torque?

Dipole in Electric Field



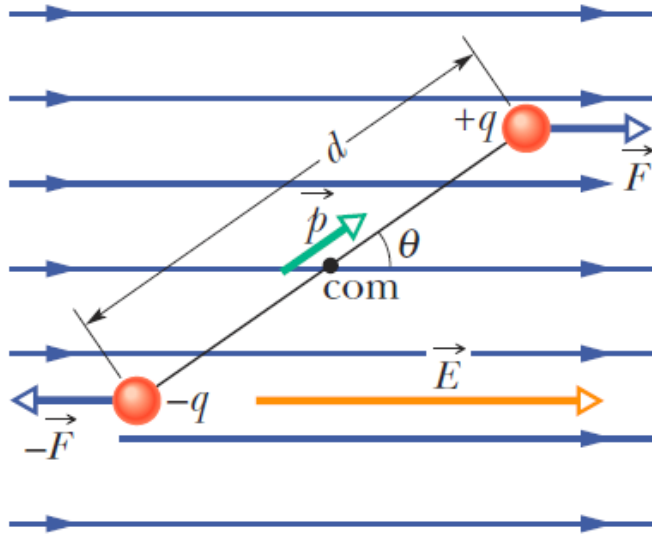
If we define the potential energy $U = 0$ to be at $\theta = 90^\circ$, then for arbitrary θ ,

$$\begin{aligned} U = -W &= -[qEx \cos \theta + qE(d - x) \cos \theta] \\ &= -qEd \cos \theta \end{aligned}$$

or $U = -\mathbf{p} \cdot \mathbf{E}$

What does this potential form imply?

Dipole in Electric Field



How does the dipole behave in the field?

$$\tau = \mathbf{p} \times \mathbf{E}$$

So, it should start rotating, and then...

It oscillates!

Keep in mind: energy conservation!

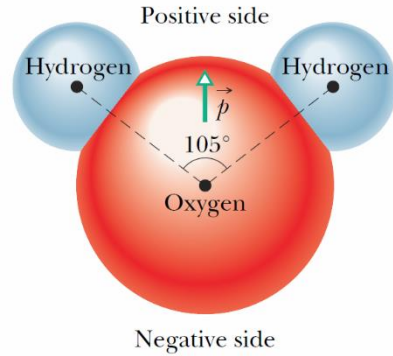
$$\text{Energy} = -pE \cos \theta + 0 = -pE + \frac{1}{2} I \omega^2$$

$$\omega = \sqrt{2pE(1 - \cos \theta) / I}$$

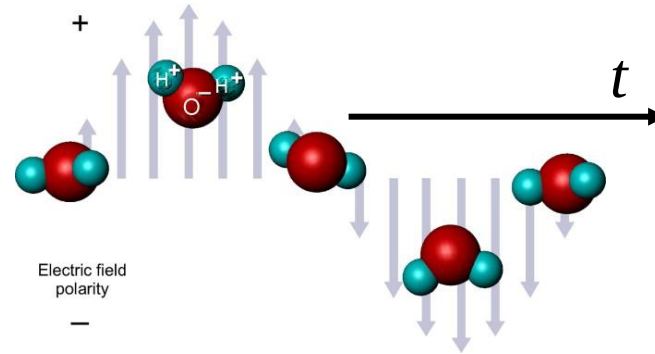
Is the period of oscillation equal to $2\pi/\omega$? Why?

How to find the period of oscillation?

Example of Electric Dipole: Water Molecules



Water molecule is a dipole.



It oscillates in an *ac* electric field.



And...we have microwave oven!